

Optimizing the Kinetic Database Algorithm for AKMC, with Applications in Studying the Kinetics of Catalyst Regeneration

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Abstract

Adaptive Kinetic Monte Carlo (AKMC) is a powerful method for simulating long-timescale chemical reactions, but its high computational cost, driven by on-the-fly saddle point searches, remains a significant bottleneck. This work analyzes strategies to optimize AKMC by focusing on the **Kinetic Database (KDB)**, an acceleration tool that reuses previously discovered reaction mechanisms via a graph-based mapping algorithm. To assess the performance of this approach, we performed a computational analysis on a dataset of 1,514 atomic structures, comparing the speed and accuracy of the graph method against a brute-force geometric search. The results show that while the graph algorithm is significantly faster, it has a measurable failure rate (~2% in this study) due to inherent problems, such as its sensitivity to fixed distance cutoffs for defining atomic connectivity. A further comparison to alternative geometric (ICP) and deep learning (RPM-Net) algorithms highlights the fundamental trade-offs between topological and geometric search methods. This quantitative analysis of the KDB algorithm's performance provides a critical benchmark for developing more robust and efficient simulation tools for the computational design of new materials.

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1. Introduction

Regenerable bifunctional catalysts such as Pt/K-MFI¹ play a critical role in industrial hydrocarbon synthesis; however, they are prone to deactivation through coke buildup and metal sintering. Furthermore, current regeneration practices are largely empirical, as the atomistic processes governing catalyst reactivation are not well understood, limiting the ability to optimize these protocols. While the project is motivated by understanding this reaction, this paper will not present the simulation results of that system due to the extensive computational time required. Rather, this paper will document and analyze the computational techniques used to optimize the core simulation technique, Adaptive Kinetic Monte Carlo (AKMC), focusing on the implementation and inherent challenges of its associated algorithms.

We aim to computationally model the kinetics, specifically using AKMC³ to simulate the catalyst's evolution, and the confluence of these methods will generate complex datasets that require a powerful platform for analysis. As a result, the Kinetic Database (KDB)—a significant tool- is ideal for data-management, enabling the crucial correlation between simulation and experiment to uncover new mechanistic insights.

1.1 Kinetic Monte Carlo (KMC) and Adaptive Kinetic Monte Carlo (AKMC)

Modeling the long-timescale kinetics of chemical and material systems is often challenging, as these are often governed by rare events that are practically inaccessible by Molecular Dynamics (MD) due to vast differences in the MD timestep and experimental timescales.. However, there have been several approaches to counteract this issue, one such approach being the Kinetic Monte Carlo (KMC)⁶. This method can model the state-to-state kinetics of such systems, but it requires that all possible reaction mechanisms and their rates are known beforehand. This strict requirement limits the applicability of KMC to only the simplest of processes where every potential event can be predetermined⁴. To overcome this limitation, the

Adaptive Kinetic Monte Carlo (AKMC) was developed with the ability to obviate the breadth of previously-known information. Essentially, the AKMC relies on knowing the atomic configuration (the positions of the atoms) of the starting point or reactant state. Thus, while the AKMC would still necessitate information regarding the reactant state, this would be an obvious improvement over the KMC in regards to information known *a priori*. After receiving this information, the AKMC determines the rate and exit processes for each state involved in the simulation and using an min-mode following algorithm, the AKMC detects saddle points—a point on an energy surface that is a maximum along the reaction path but a minimum in all other directions, representing the transition state between two stable states—in a given energy plane.

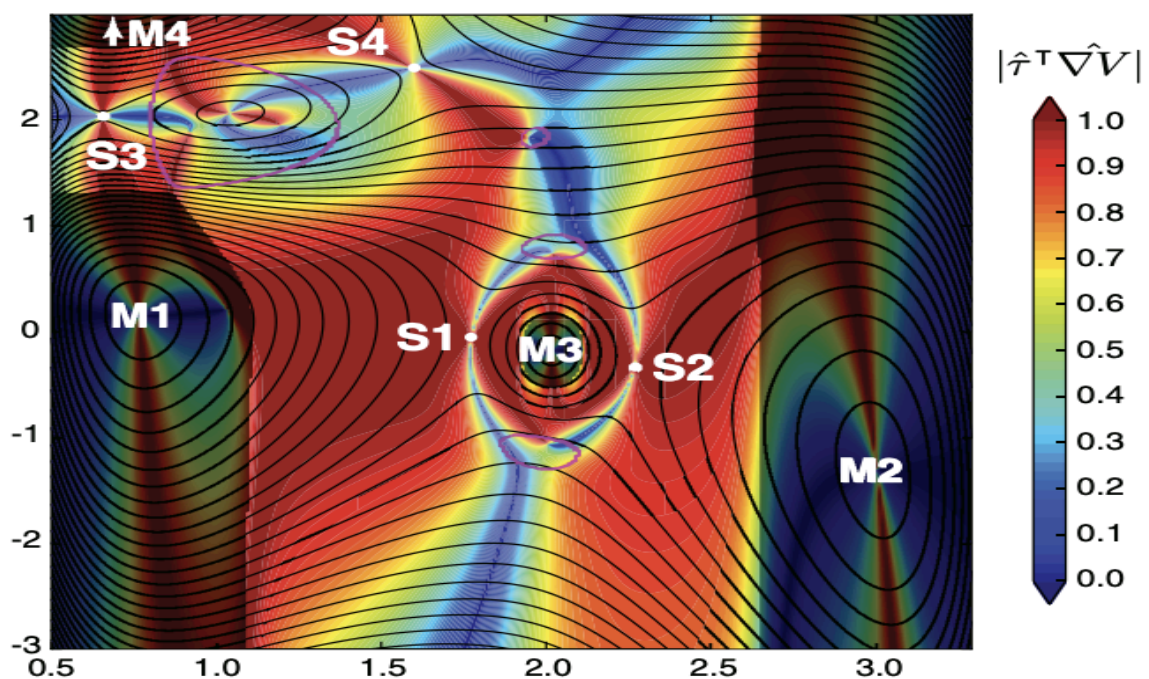


FIG. 1 in this figure, the points labelled with an S indicate saddle points. This indicates a shift from a stable or metastable state (M1, M2, M3, and M4) to another.

To ensure that all relevant saddle points have been visited, the AKMC model uses a confidence parameter,

$$C = 1 - 1/(\alpha N r),$$

where N_r represents the number of sequential searches that find previously discovered energy processes and α represents a sort of rarity ratio between the most difficult energy process to detect and the easiest one. That being said, the AKMC does also carry certain disadvantages in regards to its computational efficiency, demanding a larger sum of resources to search for processes in each new state visited (Terrell et al.). Specifically, the AKMC is made up of many KMC steps, each KMC step requires thousands of saddle searches, and each saddle search requires hundreds of energy/force calculations from electronic structure methods, like DFT, and each of those calculations take hours

1.2 Kinetic Database (KDB)

Standard AKMC is computationally expensive because it must perform many random searches for reaction pathways (saddle points) in every new state it visits. While methods like saddle point recycling help, they are limited because they only reuse information from the single, immediately preceding state of the simulation. The Kinetic Database (KDB) was developed to overcome these limitations. Essentially, the KDB stores and acts as a type of memory for the AKMC, enabling it to become more efficient. When a new reaction is found, the KDB stores a "minimal representation" of that process. This minimal representation of the hop is then stored in the database as three separate configurations: the start of the process (reactant), the middle of the transition (saddle), and the end of the process (product). Subsequently, when the simulation enters a new state, it queries the KDB to find potential reactions rather than starting random searches from scratch. This implementation of KDB helps AKMC avoid doing redundant saddle searches.

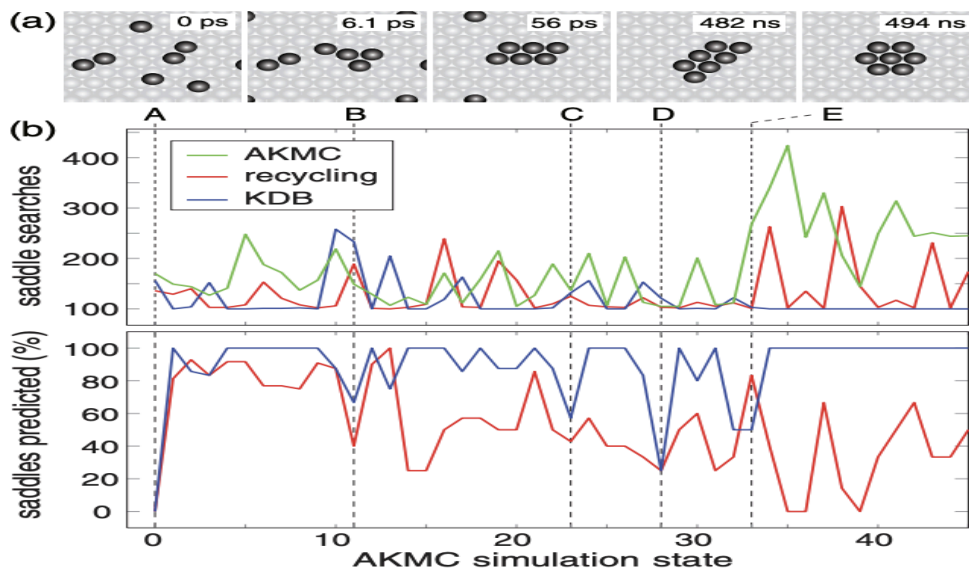


FIG. 2 KDB driven trajectories had on average 1.4% of the computational cost of standard AKMC simulations

2. Methodology

The selection of Adaptive Kinetic Monte Carlo (AKMC) over standard Kinetic Monte Carlo (KMC) for this project, despite its inefficiency relative to KMC, is a direct consequence of KMC's main limitation: its requirement that all reaction pathways be known *a priori*, a condition not met by the complex Pt/K-MFI catalyst system. While both methods can access long timescales, AKMC was chosen for its ability to discover new and unexpected reaction mechanisms during the simulation, which is essential for developing a true mechanistic understanding of this complex catalyst. While AKMC's computational cost exceeded that of the KMC, this is lowered by implementing KDB. However, this method has its own problems, specifically in the subgraph isomorphism algorithm³ used to query the database, which unfortunately tends to struggle to effectively handle highly symmetric structures and disconnected atom clusters. Unfortunately, this project will not document the catalysis simulation, as it would take around six months for a complete simulation of the process, but this will document the strategies used to optimize the speed of the currently used algorithm. To address the primary issue associated with AKMC, its inherent inefficiency, we decided to use and optimize KDB in hopes of reducing the time required to output certain mappings. To test our processes, we used a dataset

consisting of 1514 examples and documented important values such as the total time taken to insert and query a given example in seconds, the time taken to just query in seconds, and the number of matches found. Interestingly, there were several instances where we got zero matches despite just having inserted the structure into the KDB, and to understand why, we needed to investigate the inherent limitations of the graph method so that we can establish it as a benchmark for developing more efficient query methods.

2.1 The Inherent Problems of the Current Mapping Algorithm

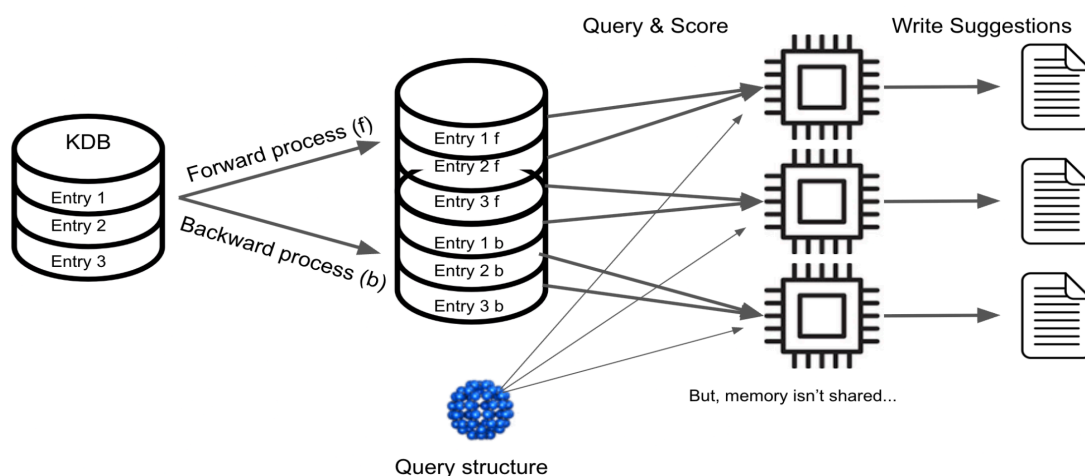


FIG. 3 illustrates the KDB insertion and querying process

While the graph-based mapping algorithm is powerful, it has several inherent problems that can lead to failed matches or computational bottlenecks. These issues stem from the way atomic structures are converted into graphs and from the nature of the graph-matching problem itself. One such inherent problem of the current mapping algorithm occurs when the algorithm generates a graph by drawing an "edge" between two atoms only if the distance between them is less than a fixed neighbor cutoff value,

$$\text{cutoff} = (r1 + r2) * (1 + nf),$$

where r_1 and r_2 represent two atomic radii and nf is a “neighbor fudge” parameter that scales the bonding distance to create a buffer, essentially extending the maximum distance permitted to have a formed edge. However, this could lead to problems wherein a bond length in the KDB structure is just *below* this cutoff, while the corresponding bond in the query structure is just above it. This could lead to certain discrepancies, invalidating potential mapping efforts made by the AKMC using KDB.

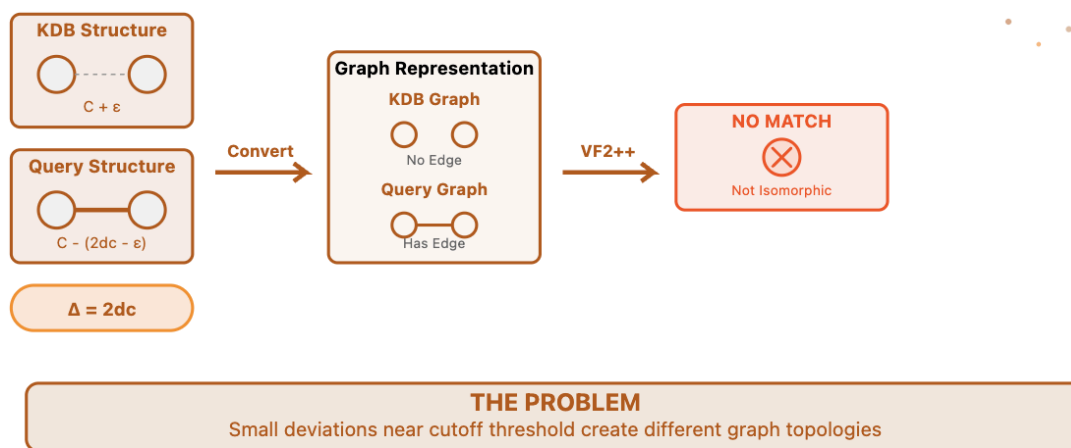


FIG. 4 visually depicts IPGM 1

Another issue arising from the current process lies in its consideration of small cells, occurring when a KDB structure is so large that it wraps around the periodic boundary conditions of its original unit cell, but is then compared to a query structure where that periodicity is not present. This is caused by the KDB’s storage of a minimal representation of the reaction mechanism inconsiderate of periodic boundary information. Thus, in examples wherein the example did span a periodic boundary, its graph might have fewer connections than a geometrically identical structure in a larger, non-periodic query system.

2.2 Performance Analysis of the Graph Algorithm

As AKMC does consider the Periodic Boundary Condition (PBC) we implemented a check to validate the minimum image convention. This check ensures that the longest possible bond length is less

than half of the smallest dimension, preventing the atom from incorrectly bonding to a periodic image of its neighbor instead of the primary atom within the cell. Additionally, we defined the minimum number of atoms for a successful insert into the KDB to be four rather than its previous value of one to avoid any ambiguous cases as a single atom would not have the context necessary to generate possible matches. Furthermore, we also experimented with the possibility of a dynamic mobile atom cutoff, a value designed to identify the mobile atoms from a given insert. After making these changes, we ran our performance analysis on a dataset of 1,514 examples to quantify the graph algorithm's success rate. Essentially, we inserted all of these examples into the KDB and queried from the dataset with and without the graph method, taking note of values such as the total time taken to insert and query, the total time taken to query, and the number of matches found.

3. Results and Discussion

Upon analyzing the results from the 1,514 examples, the data confirmed our hypothesis about the performance of the graph-based algorithm. The timing metrics indicated a clear improvement in efficiency when using the graph method—the graph-method was 70% faster than the brute-force geometric search. However, this improvement in speed came at a slight cost as the model was only able to correctly identify 99.6% of all unique matches when using the graph algorithm. Furthermore, we discovered that there were 31 discrepancies, as in the performance differed with and without the graph method; however, all of these discrepancies could be associated with the inherent problems of the graph algorithm.

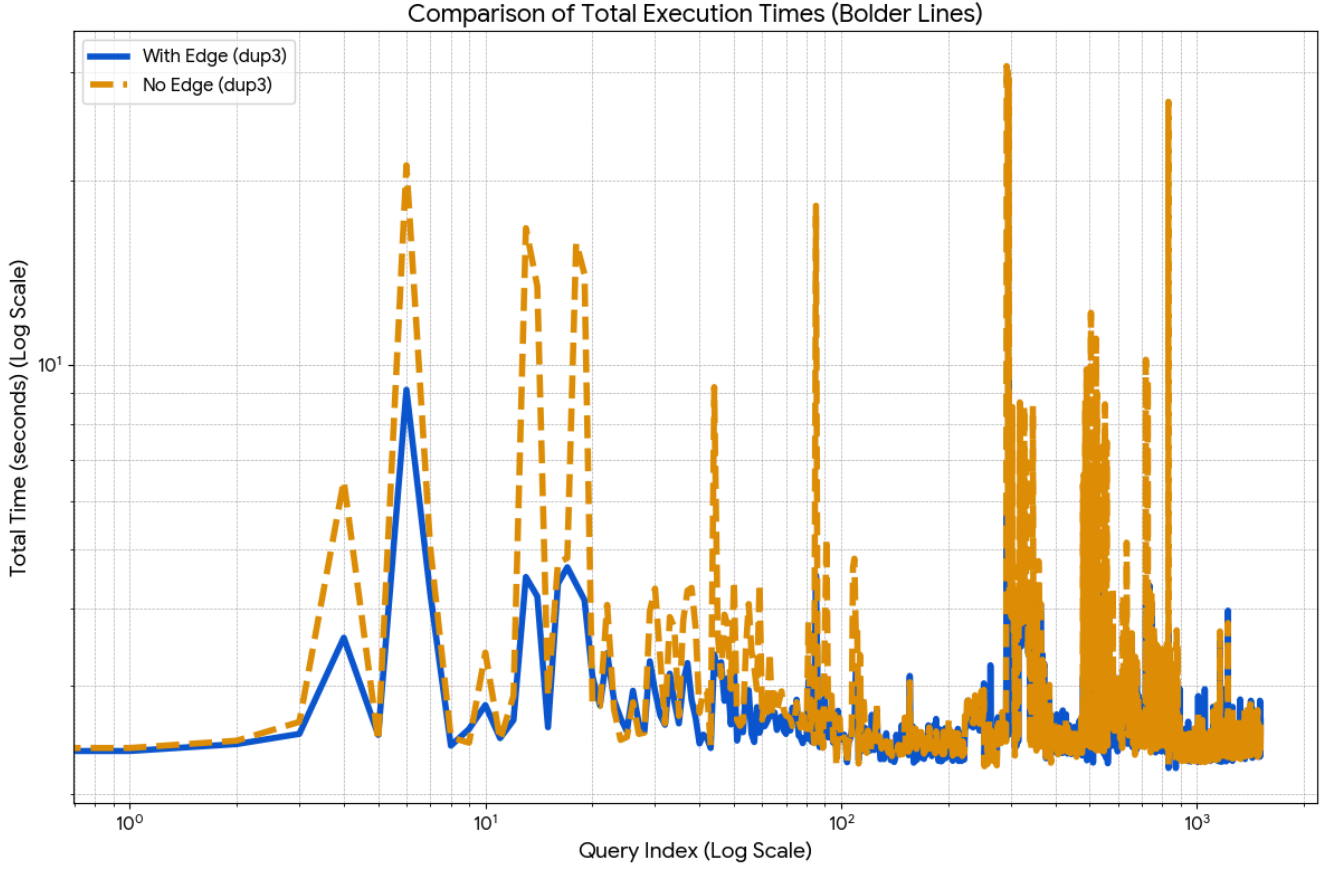


FIG. 5 illustrates the performance times of insertion and querying process with and without the graph method (The blue line depicts performance with the graph algorithm and the orange line represents the performance without the graph algorithm)

After getting the graph-algorithm to a state where the only mistakes it makes are inherent issues with the approach and seemingly no other issues exist in the method, we have begun to work on alternative methods, and we will compare these methods to the refined graph-algorithm.

3.1 Iterative Closest Point (ICP)

The first approach we have decided to compare the graph-method to is the Iterative Closest Point (ICP). This algorithm aligns two point clouds by repeatedly finding the closest point in the target cloud for each point in the source cloud and calculating a transformation that minimizes the distance between

these pairs. Essentially, this algorithm views distance as $\|x-y\|$, and then minimizes the sum of the squared distances between each point where

$$\text{sum} = \sum \|(Rx + t) - y\|^2$$

in which R represents the rotation matrix and t is the translation vector. Due to it being a purely geometric approach, ICP would not be susceptible to the aforementioned missing edge problem. However, this immunity comes at a steep price with the quintessential ICP algorithm prioritizing proximity over bonding connectivity, an issue that could lead to physically incorrect matches. Furthermore, ICP typically requires a good initial guess for the alignment to converge correctly—a requirement not mandatory for the current graph approach. That being said, ICP could also lead to a significant improvement in efficiency relative to the current graph-based approach.

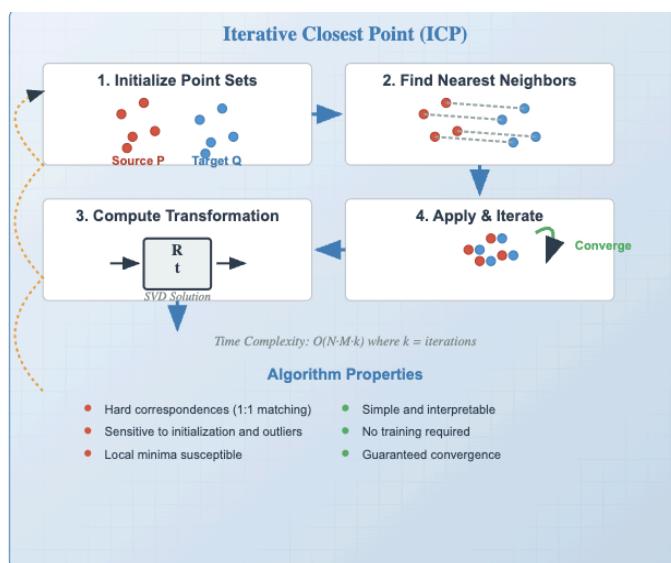


FIG. 6 illustrates the ICP's process

3.2 Robust Point Matching (RPM-Net)

The second approach we have been considering is RPM-Net. Instead of converting atomic structures into graphs, RPM-Net uses a neural network to process point clouds and computes a feature

vector for each atom. This is opposed to the graph method, which represents atoms simply as nodes connected by edges based on a strict distance cutoff. By learning features from local geometry, RPM-Net can encode more subtle information about the environment of an atom than the binary "connected" or "not connected" information in the graph model, and is therefore intrinsically robust to small geometric distortions that will ruin the topology of the graph and create a "missing edge" problem.

The basic methodological distinction is in how the two approaches find correspondences. The graph approach searches for a "hard," one-to-one correspondence by calculating the graph-subgraph isomorphism problem; a subgraph is either topologically identical to the KDB entry or it is not. RPM-Net, in contrast, calculates "soft correspondences" by building a probability matrix where each entry represents the probability of a match between two points in the two clouds ². This matrix is then normalized by a Sinkhorn normalization layer, which helps the network learn to ignore outliers—a property that the rigid isomorphism search lacks. This "soft" approach allows RPM-Net to handle noise and partial matches, whereas an additional or missing atom would cause the graph isomorphism to fail.

Additionally, once the graph method gets a valid topological match, it employs geometric alignment not only to score the match but also to validate the match. For RPM-Net, the final matrix of soft correspondences directly affects the solution by being used as weights in a weighted least-squares calculation for calculating the singular best rotation and translation that best aligns the point clouds. So, while the graph method is a combinatorial search for a precise topological pattern, RPM-Net is a holistic, data-driven optimization that finds the best geometric fit based on learned features. The implementation of this approach could lead to optimizations in accuracy compared to the current graph-algorithm at the cost of a reduced efficiency.

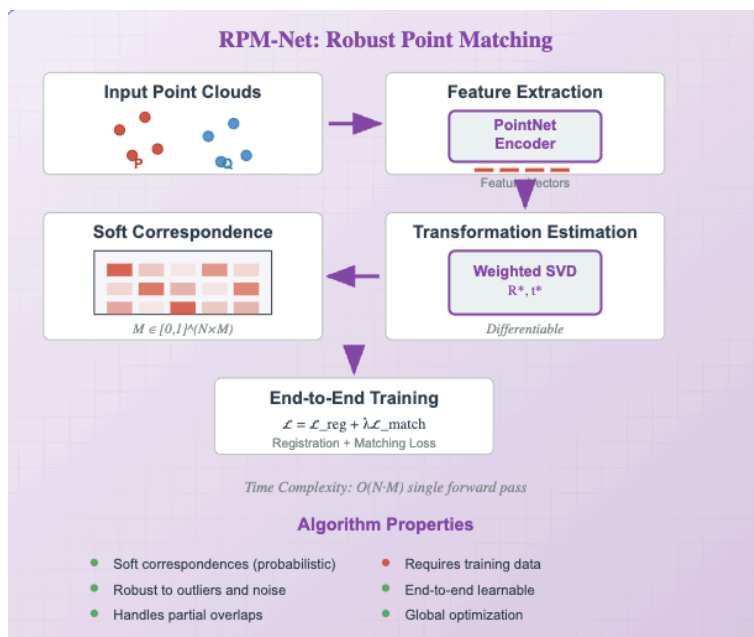


FIG. 7 illustrates the RPM-Net's process

4. Conclusion

In this study, we successfully demonstrated the benefits of implementing the graph-algorithm into KDB and we have also measured its failure rate, providing insight into the limitations of our current approach and serving as a benchmark for future optimization efforts. Furthermore, through optimizing the KDB, we have enabled a method wherein we could potentially mitigate the time taken for the AKMC to produce a simulation of our desired catalysis reaction.

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